

Approaches to HL-LHC Prompt vs Non-Prompt Muon Track Discrimination

Sergei Uzunian

Advisors: Jahred Adelman, Kevin Sedlaczek

August 18 2026

Introduction. The Large Hadron Collider (LHC) is expected to begin Run 5 with pileup $\langle\mu\rangle \approx 200$ around the late 2030's following the High Luminosity (HL) upgrade. The significant increase in simultaneous proton-to-proton collisions complicates the problem of distinguishing between prompt and non-prompt lepton tracks [Figure 1], including muons. Prompt leptons emerge from the decays of electroweak and Higgs bosons, whereas non-prompt muons tend to arise in hadron decays.

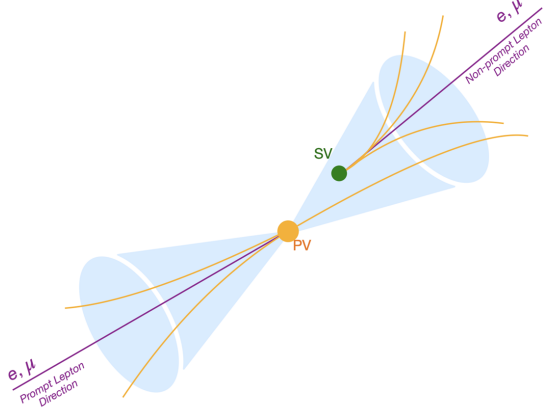


Figure 1. Illustration of prompt and non-prompt leptons, reproduced from [1].

Transformer-based machine learning approaches have been successful in prompt vs non-prompt lepton track discrimination for previous LHC runs 2 and 3 [1]. As a part of US-ATLAS SUPER, I have developed and compared the viability of Python-written XGBoost library BDT and PyTorch Neural Network approaches for muon track discrimination operating on inputs from inner detector track reconstructed muon and nearby track kinematics and impact parameter information. While more sophisticated than standard isolation scalar cut-based discrimination (wherein relative isolation for candidate muon is summed over a ΔR cone and scanned by threshold), these approaches remain simpler than transformer-based architecture and appear to offer promising preliminary results at relatively small sample size, outperforming the isolation-cut baseline as seen from Receiver-Operator Characteristic (ROC) curves.

Furthermore, I have done optimization work on a heuristic-motivated if-else table developed by the

NIU ATLAS group for a more refined cut-based isolation approach to prompt vs non-prompt muon track discrimination which defines a $|\Delta z_0 \sin \theta|$ window (adding additional constraints to a ΔR cone) for a nearby track's inclusion to a candidate muon's isolation scalar calculation on the basis of the nearby track's p_T and η values.

Methodology. Training data was simulated for HL-LHC $\langle\mu\rangle = 200$ with a signal sample of single prompt muons with pileup and background of from b-hadron decays in HH(4b) events. Signal consisted of 9291 events used with 900 tracks/event (and one muon/event), and background with 9999 events used with $\sim 1,970$ tracks/event.

Each event stores list of reconstructed inner-detector tracks including per-track kinematics p_T , η , ϕ and impact parameter values $z_0 \sin \theta$ (longitudinal), d_0 (transverse).

For the machine learning discrimination approaches, each candidate is a reconstructed muon. The candidate's feature vector consists of the muon's own p_T , η , $|z_0 \sin \theta|$ and d_0 together with a block of up to $K = 10$ neighboring tracks from a cone defined by ΔR and $|\Delta z_0 \sin \theta|$ about the candidate, each neighbor carrying its d_0 , p_T , $\Delta \eta$, ΔR and $\Delta z_0 \sin \theta$ relative to the muon. Since the number of tracks inside the cone varies between events, the neighbor slots are filled in two orderings (ΔR -ascending and p_T -descending), with unfilled slots NaN-padded for the BDT (handled natively by the XGBoost library) and zero-padded for the Neural Network. Tracks below 1 GeV are dropped.

Two kinds of reweighting are combined into one per-muon training weight. The first is kinematic: since the samples differ in their muon (p_T , η) spectra, a gradient-boosted reweighter (fit on the training set only) weights the background so that its muon (p_T , η) matches signal, preventing separation on sample kinematics instead of isolation. The second corrects class imbalance, upweighting each signal muon by $n_{\text{bkg}}/n_{\text{sig}} \approx 62$ ($\sim 465\text{k}$ background vs $\sim 7.5\text{k}$ signal training muons).

The data is split 80/20 by event into training and holdout sets, and all quoted performance is the holdout AUC (ROC area under curve) computed on the 20% of muons the model never saw, with the training/holdout AUC gap used as the overfitting benchmark. Youden's $J = \text{TPR} - \text{FPR}$,

maximized over the classification score threshold, provides a distinguished operating point weighting correct identification of prompt muons and rejection of non-prompt muons equally.

The Neural Network is a PyTorch multilayer perceptron of $105 \rightarrow 15 \rightarrow 15 \rightarrow 1$ with ReLU hidden layers and $\sim 1,850$ parameters, trained with weighted cross-entropy and the Adam optimizer for 21 epochs on the same inputs, preprocessing and split as the BDT. The cut-based baseline computes relative track isolation $\sum p_T^{\text{cone}}/p_T^\mu$ over the same cone, scanned over ΔR and the $|\Delta z_0 \sin \theta|$ window for the best-performing isolation value, serving as a proxy for any cut-based isolation tool.

Results. Following my advisor’s guidance, the analysis is restricted to a physically motivated cone $\Delta R \leq 0.5$ (the jet scale), at which the BDT reaches a holdout AUC of 0.8822 [Figure 2], against 0.8604 for the Neural Network and 0.7510 for the best cut-based isolation.

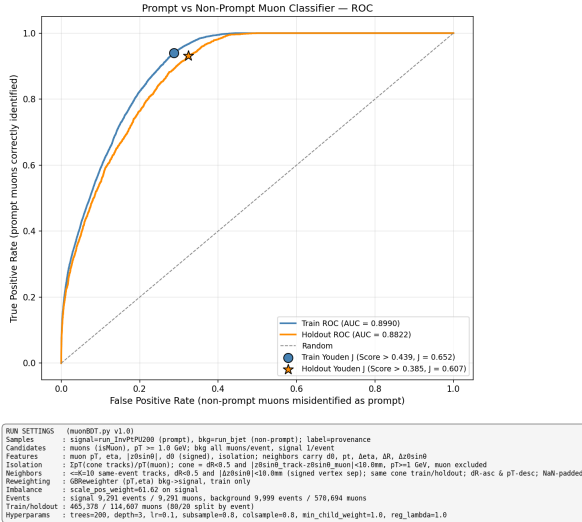


Figure 2. ROC curve of the best BDT at the physical cone $\Delta R \leq 0.5$ (holdout AUC = 0.8822).

Feature importances [Figure 3] show that most of the discrimination is carried by the surrounding track activity rather than by the muon’s own parameters. Consistent with this, the muon’s own d_0 gives a small gain (0.8822 with vs 0.8744 without), the neighboring-track d_0 is negligible (0.8744 vs 0.8760 with d_0 removed everywhere), and the optional isolation scalar and zeta ordering ($\zeta = \sqrt{(20 \Delta R)^2 + (\Delta z_0 \sin \theta)^2}$, a combined angular-longitudinal distance metric tested as a third neighbor ordering during development) add +0.000 and +0.0004 respectively, being already captured by the neighbor-track features.

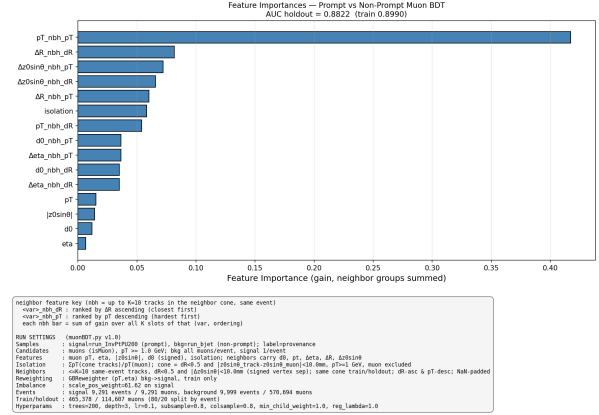


Figure 3. Feature importances of the best BDT, with neighbor-track groups summed over all K slots.

The cut-based isolation was scanned over the cone radius and the longitudinal window [Figure 4], reaching its best AUC of 0.7510 at $\Delta R < 0.4$ with the “auto” window taken from the if-else Table 1, as opposed to any single fixed cut.

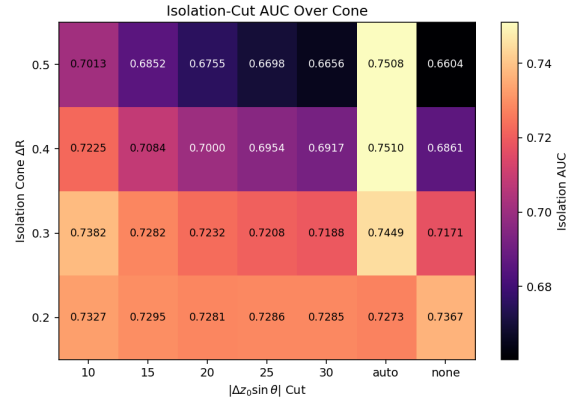


Figure 4. Isolation-cut AUC over the cone radius and the $|\Delta z_0 \sin \theta|$ window.

For the table optimization mentioned in the introduction, the table’s p_T and η bin values were scaled and the cone rescanned for each variant [Figure 5]. Halving or doubling the η bin values moves the AUC by only less than 0.007 in either direction, whereas dividing the p_T bin values improves performance up to a maximum at one third, beyond which the discrimination performance falls away again. The best isolation AUC accordingly rises from 0.7510 to 0.7740, obtained with the table shown in Table 1. Since dividing the p_T bin values pushes each track into a lower table row, the improvement corresponds to a systematically wider $|\Delta z_0 \sin \theta|$ window than the original prescription.

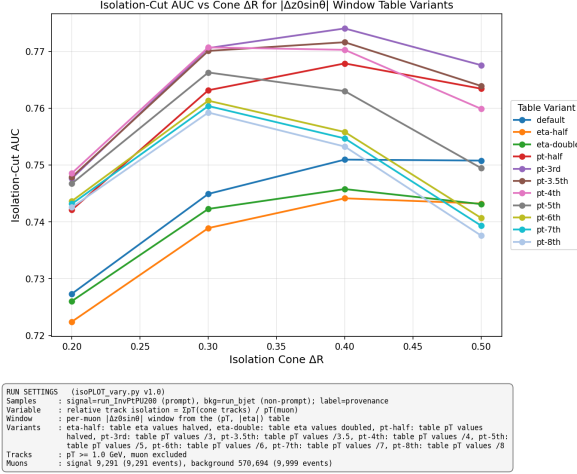


Figure 5. Isolation-cut AUC versus cone ΔR for variants of the $|\Delta z_0 \sin \theta|$ window table.

p_T [GeV]	$ \eta < 1$	$1 < \eta < 2$	$ \eta > 2$
1.00–1.33	2.0	1.6	1.0
1.33–1.67	3.3	2.5	1.6
1.67–2.00	4.2	3.3	2.5
2.00–2.33	5.4	5.2	4.2
2.33–3.33	6.0	5.2	4.2
3.33–6.67	6.0	5.5	4.2
6.67–10.00	6.0	6.6	5.4
> 10.00	6.0	8.5	8.5

Table 1. Best performing $|\Delta z_0 \sin \theta|$ window table [mm], with the original p_T bin values divided by 3.

Evaluated on the same holdout muons, the BDT outperforms both the Neural Network and the cut-based isolation across the whole efficiency range [Figure 7], attaining Youden’s $J = 0.607$ at its chosen operating point against 0.562 and 0.439 respectively. That the Neural Network trails the BDT by only 0.022 in AUC despite the padding disadvantage and limited signal statistics suggests it may still outperform once better optimized.

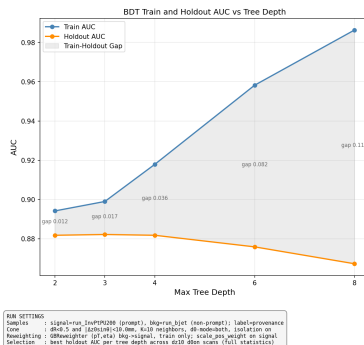


Figure 6. Train–holdout AUC gap as a function of tree depth.

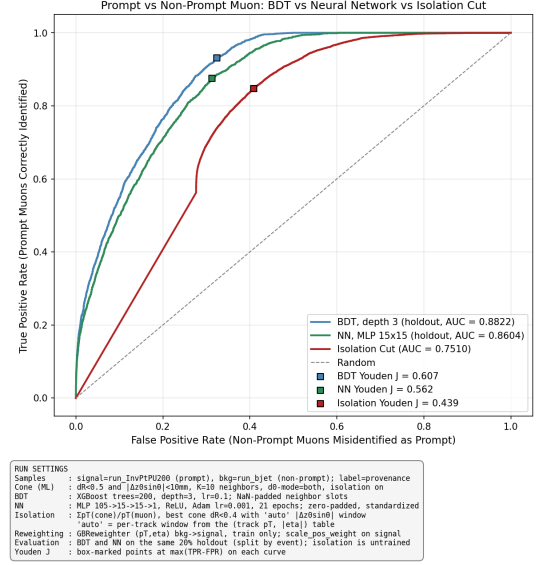


Figure 7. Holdout ROC curves of the BDT, the Neural Network, and the cut-based isolation.

The Table 1. optimizations occurred following a question asked by Matteo Baucé at a presentation I gave at the CERN Isolation and Fakes Forum (IFF) on July 27th 2026.

Challenges. Development required keeping track of collected data across many interacting settings (cone cuts, d_0 modes, reweighting, network architecture), where the output of two runs is indistinguishable without a record of the settings which produced it. This was addressed by the addition of a standardized run-settings box added to every plot, with every run also writing a summary file and a settings-encoded output directory.

Scanning the parameter space required many hundreds of runs (over 500 BDT runs across the cone, d_0 and hyperparameter scans, with full statistics, and often runs taking hours each). This was addressed with batch processing on the NIU METIS cluster, employing reusable and tunable PBS scripts which can build the features once and reuse them across configurations.

Another issue was overfitting; it is well-known that machine learning models can memorize their training data, and early BDTs reached training AUC near 1.0. This was addressed with the 80/20 split by event and reporting only holdout AUC, after which lowering tree depth was found to narrow the train–holdout gap [Figure 6], the holdout AUC itself falling away beyond depth 3.

A subtler difficulty is that the holdout AUC rises with cone ΔR and never plateaus [Figure 8], reaching ≈ 0.97 once the cone spans the whole event, which cannot reflect genuine isolation physics. The current hypothesis is that widening the cone fills more of the K neighbor slots, with the mean filled slots rising from ~ 2 at $\Delta R = 0.2$ to

all $K = 10$ by $\Delta R \approx 1.5$ [Figure 9]. Hence, a wider cone supplies more neighboring track information. This was addressed by restricting the analysis to $\Delta R \leq 0.5$ “physical” approximation.

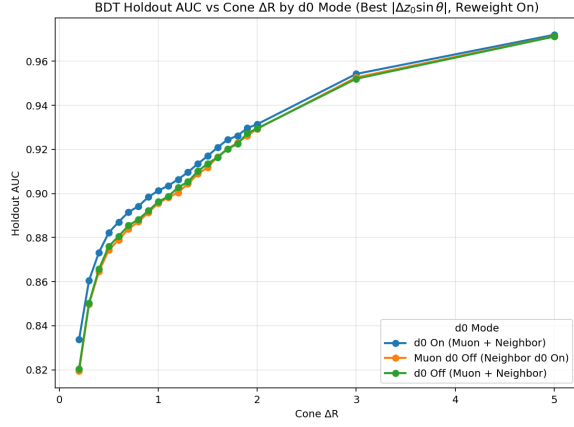


Figure 8. Holdout AUC versus cone radius ΔR for the three d_0 feature configurations.

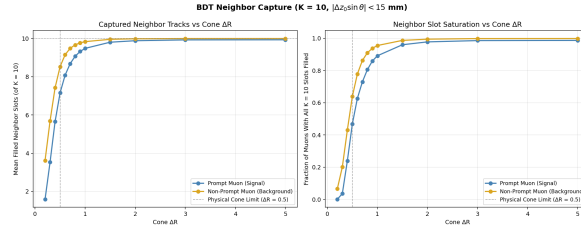


Figure 9. Mean filled neighbor slots and slot saturation versus cone ΔR for $K = 10$.

Finally, with a small sample data size of only $\sim 7,400$ simulated signal muons available for training, the Neural Network overfits within a few epochs, and the holdout loss turned upward while the training loss kept falling [Figure 10]. This was addressed by tracking holdout loss and AUC at every epoch and employing a small network of $\sim 1,850$ parameters, taking the epoch of maximum holdout AUC as the stopping point.

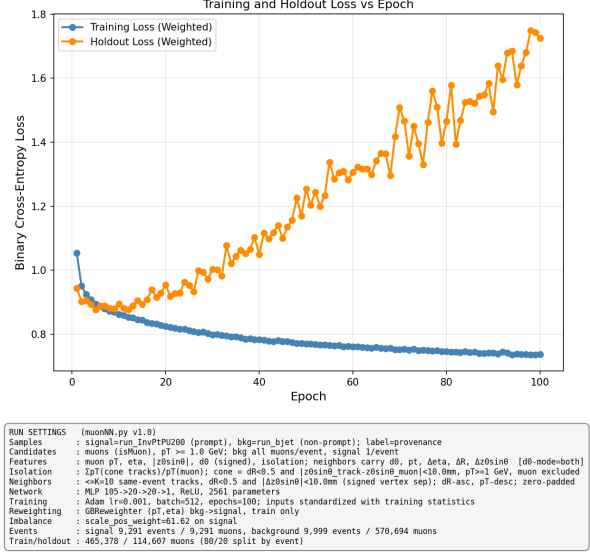


Figure 10. Training and holdout loss versus epoch for the Neural Network.

Future Directions. Arguably the clearest limitation of the present has been the signal sample itself, in which the prompt muon is added directly without a simulated decay event.

The next steps would involve training on simulated samples in which prompt muons emerge from actual W , Z or Higgs decays (as opposed to single muon injections in the current approach), paired with a greater sample size, as well as increasing the number of neighboring tracks beyond $k = 10$. With only $\sim 7,500$ signal muons available for training, both models are statistics-limited, and a larger sample would permit deeper trees and larger networks before overfitting sets in.

Additionally, the present analysis uses inner detector track information only. Incorporating calorimeter data, such as the transverse energy deposited in a cone around the candidate muon, would supply an additional independent handle on the surrounding track activity, and is considered standard in ATLAS isolation tools.

With these in place, the goal is to develop the BDT and Neural Network approaches to experimentally-useful discrimination levels, evaluated as non-prompt rejection at fixed prompt efficiency working points rather than by AUC alone, for eventual comparison against the isolation taggers used by ATLAS.

References

- [1] ATLAS Collaboration, “Development and performance of a transformer-based lepton isolation algorithm for the LHC Run 3 with the ATLAS experiment,” ATLAS Paper Draft MUON-2024-01, draft version 1.0, 7 August 2026. Unpublished; target journal *JINST*. Supporting internal note: <https://cds.cern.ch/record/2952919>.